Adaptive Budget Scheduling for Constrained RL Deployment via Energy Composition

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Abstract

Deploying RL policies under cumulative operational constraints (maintenance budgets, duty-cycle limits) typically requires retraining for each constraint specification. We propose a closed-form penalty weight based on the *urgency ratio*—remaining budget divided by remaining time—that composes post-hoc with pre-trained policies via energy addition or gradient correction, requiring zero retraining. A single parameter η controls risk aversion; a single policy serves arbitrary budget levels instantly. On wind farm yaw control, the schedule outperforms an oracle-tuned constant penalty (98.9% vs. 96.7% power retention) and a nonlinear fatigue surrogate achieves 6.8× structural fatigue reduction. We identify the *action-to-cost coupling strength* as the scope condition: post-hoc correction works for action-dependent costs (wind farm) but partially fails for state-dependent costs (Safety Gym), where a learned cost critic Q_c reduces cost by 20–36% but cannot satisfy tight budgets.

1 Introduction

Consider a wind farm operator who trains a reinforcement learning (RL) controller to maximize power production. Six months after deployment, a bearing inspection reveals that one turbine needs restricted yaw travel: it may operate at negative yaw angles for at most 15 hours per year to limit structural fatigue. Retraining the controller is expensive and disruptive. The operator needs a way to impose this new cumulative constraint on the *existing* policy, instantly and without retraining.

This scenario is ubiquitous: cumulative operational constraints arise whenever an agent must limit total time in a costly regime—wind turbines with load budgets [van Dijk et al., 2025], robots with joint-wear cycles, grid batteries with cycle-life limits, and advertisers with spending caps [Balseiro et al., 2024]. Unlike instantaneous constraints, these are *lifetime budgets*: the agent must decide *when* to spend its limited capacity for risky actions.

Inspired by optimal trade execution [Almgren and Chriss, 2001], where a trader paces liquidation based on remaining inventory vs. remaining time, we propose an *urgency ratio* that adapts constraint strength to budget state.

Existing approaches do not solve this: **CMDP methods** [Altman, 1999, Stooke et al., 2020, Sootla et al., 2022] require retraining for each budget specification. **Post-hoc composition** [Urain et al., 2023, Zhang et al., 2025, Xiao et al., 2025] handles only instantaneous constraints, not cumulative budgets spanning an episode.

We fill this gap with three contributions: (1) a **closed-form penalty schedule** $w(u) = \exp(\eta(1/u - 1))$ derived from budget-balanced allocation, parameterized by a single risk-aversion scalar η; (2) **post-hoc composition** with pre-trained policies via energy addition (EBT actors) or gradient correction (MLP actors), enabling zero-retraining budget enforcement; (3) **empirical validation** on wind farm yaw control (6.8× fatigue reduction, 98.9% power) and Safety Gymnasium (diagnostic of state-dependent cost limitations).

2 Background

2.1 Constrained MDPs and Lagrangian Relaxation

A CMDP [Altman, 1999] augments the MDP objective with cost constraints: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ s.t. $\mathbb{E}_\pi[\sum_t c_t^i] \leq d^i$ for each agent i, where $c_t = c(s_t, a_t)$ is the per-step cost and d is the cost limit (following standard CMDP notation [**??**]). The standard Lagrangian approach learns a dual variable μ via primal-dual optimization [Stooke et al., 2020], requiring retraining whenever d changes.

2.2 Energy-Based RL and Post-Hoc Composition

Energy-Based Transformer (EBT) actors [Gladstone et al., 2025] generate actions by gradient descent on a learned energy $E_\phi^i(\mathbf{e}^i, a^i)$. External energy terms compose per-agent before aggregation:

$$E_{\text{total}}(a) = \frac{1}{N} \sum_{i=1}^{N} \left[E_\phi^i(\mathbf{e}^i, a^i) + \lambda_{\text{gs}} \cdot E_{\text{constraint}}^i(a^i) \right] \tag{1}$$

enabling post-hoc constraints without modifying E_ϕ [Urain et al., 2023, Zhang et al., 2025]. We extend this from instantaneous constraints to *cumulative budget constraints* via a time-varying energy weight.

3 Optimal Budget Scheduling

An agent allocates cost limit d over T steps. Per-step cost $c_t := c(s_t, a_t) \geq 0$; cumulative $C_t := \sum_{s=0}^{t-1} c_s$ (in wind fatigue, Miner's-rule damage). The *urgency ratio* captures the budget–time state:

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T - t}{T} \tag{2}$$

where ρ_t is the remaining budget fraction and τ_t the remaining time fraction. When $u > 1$, the agent has underspent; when $u < 1$, overspent. Per-agent budgets d^i yield u_t^i computed independently per agent.

Proposition 1 (Motivating analysis under idealized Boltzmann response)**.** *Consider an idealized policy that responds to penalty weight λ via a Boltzmann model: $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, where P_0 is the base risky-action probability and $\alpha > 0$ is the policy's sensitivity. Under this model, the penalty weight that maintains budget balance has a multiplicative structure determined by the urgency ratio:*

$$w^*(u) = \frac{1}{u} \tag{3}$$

Proof sketch. Budget balance requires $(T - t) \cdot P(\text{risky} \mid \lambda) = d - C_t$. Substituting the Boltzmann model, solving for λ, and computing the multiplicative weight relative to the TWAP baseline ($u = 1$) yields $w = \tau/\rho = 1/u$. The full derivation is given in Appendix A.1. □

Practical schedule. Since $1/u \to \infty$ as $u \to 0$ (budget depleted) and the Boltzmann response is an idealization that does not hold for real policy architectures, we use a parametric family that preserves the key structural properties:

$$w(u) = \exp\left(\eta \cdot \left(\frac{1}{u} - 1\right)\right) \tag{4}$$

The parameter $\eta \geq 0$ controls risk aversion: $\eta = 0$ recovers the constant-penalty baseline (no adaptation to budget state); $\eta > 0$ increases the penalty when overspent ($u < 1$) and relaxes it when underspent ($u > 1$).

Hard wall backstop. A steep exponential barrier activates when budget drops below 5%:

$$\lambda_{\text{wall}}(t) = \exp\left(k \cdot \text{ReLU}\left(1 - \frac{\rho_t}{0.05}\right)\right) \tag{5}$$

The combined weight is $\lambda(t) = \min(w(u_t) \cdot \lambda_{\text{wall}}(t),\ \lambda_{\max})$, clamped at $\lambda_{\max} = 10^6$.

Approximation quality. The practical schedule $\exp(\eta(1/u - 1))$ matches $u^{-\eta}$ within 5% for $u \in [0.7, 1.3]$; the approximation quality is detailed in Appendix Figure 1. Ablation results (Table 5) show that the smooth schedule alone is sufficient in practice.

Penalty landscape. The TWAP diagonal ($\rho = \tau$, where $u = 1$) separates surplus ($\lambda < 1$) from deficit ($\lambda > 1$); higher η produces steeper gradients near this boundary (Appendix Figure 2).

Note on the Boltzmann assumption. The Boltzmann response is a motivating idealization; the derivation identifies u as the natural state variable and constrains the schedule to be monotone decreasing in u, but the specific form $\exp(\eta(1/u - 1))$ is a principled parametric choice rather than an optimality claim.

4 Method

The method has three components that compose independently: (1) the *budget schedule* $\lambda(t)$ from Section 3, which determines *how strictly* to enforce the constraint at each moment; (2) the *cost function* $c(s, a)$, which defines *what* is costly; and (3) the *application mechanism*, which determines *how* the penalty modifies actions. Any cost function composes with any mechanism and the same $\lambda(t)$ schedule.

4.1 Cost Functions

The simplest cost is a binary indicator $c(s, a) = \mathbf{1}[\text{yaw} < 0]$, ignoring severity. A nonlinear surrogate captures the physics: $c(s, a) = (|a|/\bar{a})^p \cdot (v(s)/v_{\text{ref}})^q$, where $p{=}2$ (quadratic yaw-fatigue) and $q{=}3$ (cubic wind-speed amplification) are physically motivated [van Dijk et al., 2025]. All cost functions compose with the same $\lambda(t)$ schedule; swapping the cost requires no retraining.

4.2 Application Mechanisms

Energy composition (EBT actors). The cost composes directly with the policy energy before action optimization: $E_{\text{total}} = \frac{1}{N}\sum_i [E^i_\phi + \lambda_{\text{gs}}\lambda^i(t)P(c(s, a^i))]$, where $P(x) = \exp(k \cdot \text{ReLU}(x)) - 1$.

Gradient correction (any actor). For standard actors, we correct the output via:

$$a_{\text{corrected}} = a_{\text{actor}} - \lambda_{\text{gs}} \cdot \lambda(t) \cdot \alpha \cdot \nabla_a\, c(s, a_{\text{actor}}) \tag{6}$$

iterated for K steps. The gradient handles nonlinear costs that action scaling cannot: $\nabla_a c$ points away from costly regions, not just toward lower magnitude.

Algorithm 1 Post-Hoc Budget-Constrained Inference

Require: Trained actor, cost limit d^i per agent, horizon T, risk aversion η, cost $c(s, a)$

$C_0^i \leftarrow 0$ for all agents i
for $t = 0, \ldots, T-1$ **do**
  Observe s_t; compute $u_t^i = \frac{(d^i - C_t)/d^i}{(T-t)/T}$, $\lambda^i(t) = \min(\exp(\eta(1/u_t^i - 1)) \cdot \lambda_{\text{wall}}^i,\ \lambda_{\max})$
  Generate a_t: {choose one mechanism}
    EBT: gradient descent on $E_\phi + \lambda_{\text{gs}} \cdot \lambda^i(t) \cdot P(c(s_t, a))$
    MLP: $a \leftarrow a_{\text{actor}} - \lambda_{\text{gs}} \cdot \lambda^i(t) \cdot \alpha \cdot \nabla_a c(s_t, a)$ for K steps
  Execute a_t, observe r_t; $c_t^i = c^i(s_t, a_t)$
  $C_{t+1}^i \leftarrow C_t + c_t^i$
end for

4.3 Algorithm

The λ computation adds 4.7 ms overhead (19.8%) on the wind farm EBT actor. The parameters $\eta \in [2, 5]$ and $\lambda_{\text{gs}} \in [0.05, 0.5]$ require mild calibration via evaluation episodes—cheaper than retraining but not zero-cost. Unlike the Lagrangian dual μ in CMDPs (learned via gradient ascent, requiring retraining), $\lambda^i(t)$ is computed in closed form, enabling instant adaptation to new budget specifications.

5 Experimental Setup

We evaluate the budget scheduling mechanism across two domains with different risky-action semantics.

5.1 Domain 1: Wind Farm Yaw Control

Environment. WindGym [Nilsen et al., 2025] with PyWake backend. We test on two layouts: a 3-turbine row and a 9-turbine 3×3 grid. Wind speed sampled from 10–14 m/s, wind direction from 225–315°, providing natural temporal variation in wake conditions.

Risky action. Negative yaw angles ($a^i < 0$) steer the rotor against the conventional direction, deflecting wakes to increase downstream power but imposing structural loads on the yawed turbine [van Dijk et al., 2025].

Budget. B = total timesteps at negative yaw per turbine per episode.

Agent. Energy-Based Transformer (EBT) actor with Soft Actor-Critic (SAC), trained unconstrained for 100k steps. The budget surrogate composes post-hoc via per-turbine energy addition at inference.

5.2 Domain 2: Safety Gymnasium Navigation

SafetyPointGoal1-v0 [Ji et al., 2023]: a point robot navigates to goals while avoiding hazard zones (c=1 on entry). Unlike the wind farm, cost is *spatial*—it depends on position, not action direction. SAC with MLP actor, trained unconstrained for 200k steps. This domain serves as a diagnostic for state-dependent costs where the action-to-cost coupling is weak.

5.3 Baselines and Ablations

Baselines: unconstrained (upper bound), hard clip (lower bound), constant penalty (η=0, static Lagrangian), and our AC schedule ($\eta \in \{2, 5\}$). Additionally, we compare against an oracle-tuned constant penalty found

Table 1: Wind farm comparison (3-turbine, budget=15/200 steps, 50 episodes). The AC schedule outperforms the oracle-tuned constant penalty (found via 15-step bisection) while requiring zero per-budget calibration.

Method	Power (MW)	% Uncon	NegYaw	Notes
Unconstrained	27.1 ± 0.6	100%	[49, 56, 61]	—
Oracle const ($k=5.86$)	26.2 ± 0.7	96.7%	[11, 13, 14]	15-step bisection
AC ($\eta=2$, gs=0.05)	26.8 ± 0.6	**98.6%**	[9, 10, 10]	Zero calibration
AC ($\eta=5$, gs=0.1)	26.9 ± 0.5	**98.9%**	[8, 9, 10]	Zero calibration

Table 2: Binary vs. nonlinear fatigue cost (3-turbine wind farm). Nonlinear: $|y/\bar{y}|^2 \cdot (v/v_{\text{ref}})^3$.

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	[66, 75, 71]	20.3
Binary $\eta=2$ (AC)	28.1	[8, 8, 10]	86.2
Nonlinear $\eta=5$ (AC)	**29.2**	[80, 81, 83]	**12.8**

via 15-step bisection search. Ablations on the wind farm test the hard guard, schedule form ($1/u^\eta$ vs. exp), and budget flexibility (Appendix).

6 Results

6.1 Wind Farm: AC Schedule vs. Oracle Baseline

Table 1 compares the AC schedule against an oracle-tuned constant penalty (15-step bisection) on the 3-turbine wind farm (50 episodes, budget=15). The AC schedule achieves 98.9% power retention with neg-yaw $[8, 9, 10]$, outperforming the oracle constant (96.7%, $[11, 13, 14]$) without any per-budget calibration.

Power retention is within 5% of unconstrained across all budget levels; the AC schedule keeps all turbines within budget while the constant penalty requires per-budget re-tuning (Appendix Figures 6–7).

6.2 Nonlinear Fatigue: Why the Cost Function Matters

Table 2 compares binary vs. nonlinear fatigue costs. The binary AC ($\eta=2$) avoids negative yaw entirely but accumulates fatigue of 86.2—*worse* than unconstrained (20.3)—because it pushes the agent toward large positive yaw angles. The nonlinear AC ($\eta=5$) uses *more* neg-yaw steps (80–83) but achieves fatigue of only 12.8 (6.8× reduction, 98% power) by permitting mild yaw at low wind speeds and restricting it at high speeds.

6.3 Ablations

The AC schedule alone keeps all turbines within budget ($[11, 11, 11]$ at $d=15$) without the hard guard; the constant penalty overshoots at $[29, 51, 58]$. The $1/u^\eta$ and $\exp(\eta(1/u-1))$ schedules perform comparably. A single policy serves budgets from 2.5% to 50% with >95% power retention from $d=10$ onward. Full ablation tables are in the Appendix.

6.4 Scope: Action-Dependent vs. State-Dependent Costs

The method succeeds when cost depends on the current action ($\nabla_a c \neq 0$). Realistic fatigue involves *state-dependent* costs—wake propagation delays decouple upstream yaw from downstream loading [van Dijk et al.,

2025]. Safety Gym [Ji et al., 2023] exhibits the same structure: position-dependent cost with momentum dynamics.

To bridge this gap, we train a cost critic $Q_c(s,a) = \mathbb{E}_\pi[\sum \gamma_c^k c_{t+k}]$ offline (MC returns, $\gamma_c = 0.99$, correlation 0.965) and modulate $\nabla_a Q_c$ via the urgency schedule. On the wind farm (strong coupling), Q_c achieves 5.7× fatigue reduction; on Safety Gym (weak coupling), 20–36% cost reduction at >97% reward, satisfying budget at $d = 40$ but not tighter (Appendix Tables 4, 3). The *coupling strength*—$\|\nabla_a Q_c\|$ relative to $\|\nabla_s Q_c\|$—determines the correction ceiling.

7 Related Work

Constrained MDPs. Lagrangian CMDP methods [Altman, 1999, **?**, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with remaining budget ρ_t (same variable we use) but still retrains the policy, so each new budget requires a new policy. TREBI [Lin et al., 2023] handles real-time budget changes via diffusion models but requires offline data and diffusion-specific training. Our method instead freezes the policy and uses a closed-form $\lambda(t)$ schedule at inference—no retraining for any budget.

Post-hoc constraint composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], control barrier functions [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle *instantaneous* constraints; none address cumulative budgets spanning an episode.

Budget pacing and optimal execution. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] provide pacing strategies. Our work adopts the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001]—remaining inventory vs. remaining time—without claiming structural equivalence to market-impact models.

8 Conclusion

We introduced a closed-form urgency-ratio schedule for enforcing cumulative budget constraints on pre-trained RL policies without retraining. On wind farm yaw control, a nonlinear fatigue surrogate achieves 6.8× fatigue reduction while retaining 98% of power; a learned cost critic Q_c discovers the same physics from data (5.7× reduction). On Safety Gym, Q_c achieves 20–36% cost reduction at >97% reward, with the action-to-cost coupling strength determining the ceiling. The method requires zero policy retraining but mild calibration of η and λ_{gs}; the Q_c extension requires offline training on logged transitions. For state-dependent costs where action-to-cost coupling is weak (momentum-dominated dynamics, wake propagation delays), future work on control barrier functions or model-predictive shielding may close the gap. Code and checkpoints are available at `[anonymized]`.

A Additional Results and Proofs

A.1 Full Proof of Proposition 1

Proof. Budget balance requires that the expected number of risky actions over the remaining horizon equals the remaining budget:

$$(T-t) \cdot P(\text{risky} \mid \lambda) = d - C_t \tag{7}$$

Substituting the Boltzmann model: $(T-t) \cdot P_0 \exp(-\alpha\lambda) = d - C_t$. Solving: $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)\,P_0}{d - C_t}$. At the TWAP baseline ($u = 1$, where $\rho = \tau$, *i.e.*, $(d - C_t)/d = (T-t)/T$), the budget balance gives $\lambda_0 = \frac{1}{\alpha} \ln \frac{T P_0}{d}$. The

multiplicative weight relative to the TWAP penalty is:

$$w = \exp\big(\alpha(\lambda - \lambda_0)\big) = \frac{(T-t)\,P_0}{d - C_t} \cdot \frac{d}{T\,P_0} = \frac{(T-t)/T}{(d-C_t)/d} = \frac{\tau}{\rho} = \frac{1}{u}$$

□

A.2 Approximation Quality

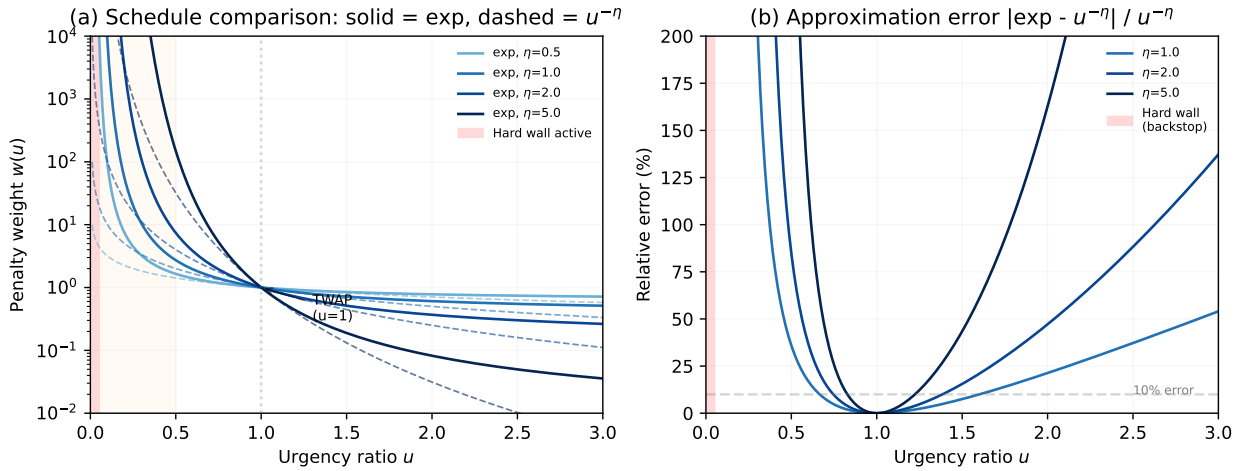

Figure 1: (a) Schedule comparison: $\exp(\eta(1/u - 1))$ (solid) vs. $u^{-\eta}$ (dashed) across the full urgency range. Red region: hard wall backstop active ($u < 0.05$). (b) Relative approximation error. The schedules agree within 10% for $u > 0.5$ (the regime where smooth pacing operates).

A.3 Penalty Landscape

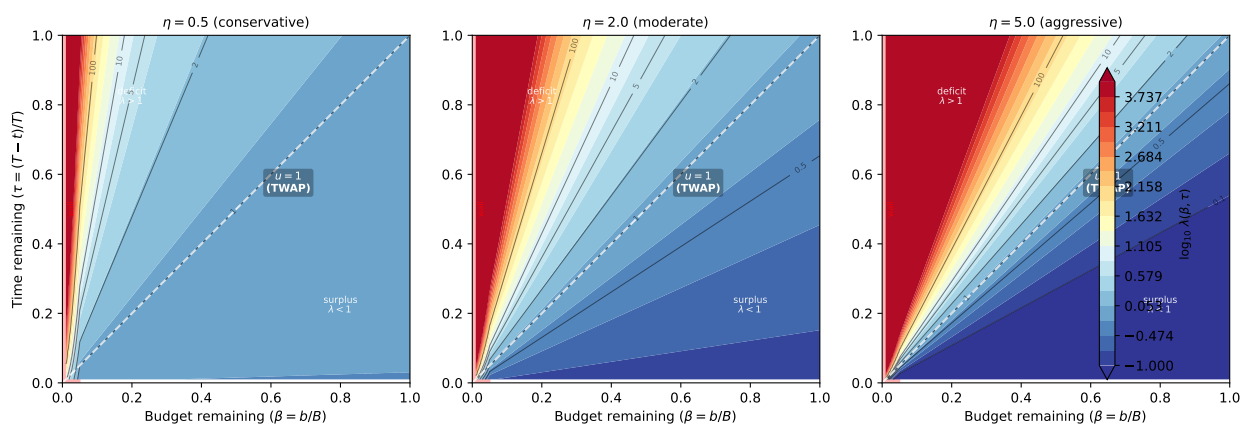

Figure 2: Penalty weight $\lambda(\rho, \tau)$ across the budget–time state space for three risk-aversion values η. The TWAP diagonal (white dashed, $u = \rho/\tau = 1$) separates surplus (blue, $\lambda < 1$) from deficit (red, $\lambda > 1$). Contour lines show penalty magnitudes. Higher η produces steeper gradients near the diagonal, concentrating budget spending into favorable conditions.

A.4 Yaw Angle Heatmaps

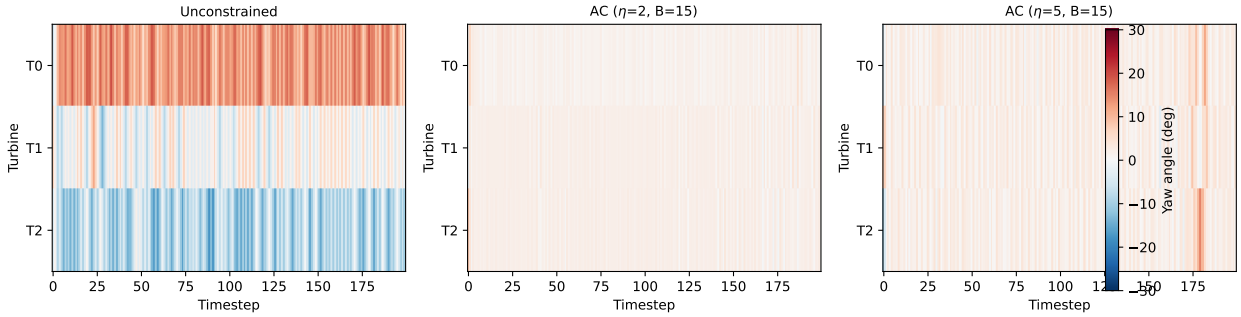

Figure 3: Yaw angle heatmaps (turbine × timestep) showing the spatial-temporal pattern. Unconstrained (left): persistent positive/negative yaw steering across all turbines. AC $\eta = 2$ (middle): nearly zero yaw—very conservative. AC $\eta = 5$ (right): mostly zero with a concentrated burst of negative yaw (blue) near the end, representing strategic budget spending.

A.5 Wind Farm Time Series and Pareto

A.6 Safety Gymnasium Diagnostic

Table 3: Safety Gymnasium diagnostic (SafetyPointGoal1-v0, 20 episodes). Q_c reduces cost 20–36% while preserving >97% reward. Budget satisfied only at $d=40$.

Method	Budget	Reward	% Uncon	Cost	OK
Unconstrained	—	26.7 ± 1.1	100%	∼50	—
Q_c gradient correction, $\eta = 5$ (fixed config)					
Q_c grad	10	26.1	98%	41	×
Q_c grad	25	26.7	100%	41	×
Q_c grad	40	26.6	100%	55	×
Best across modes (ceiling of post-hoc correction)					
Q_c sample ($\eta=5$)	10	26.2	98%	40	×
Q_c sample ($\gamma_c=0.95$)	25	26.7	99%	**29**	×
Q_c grad ($\eta=0$)	40	25.8	97%	**33**	✓

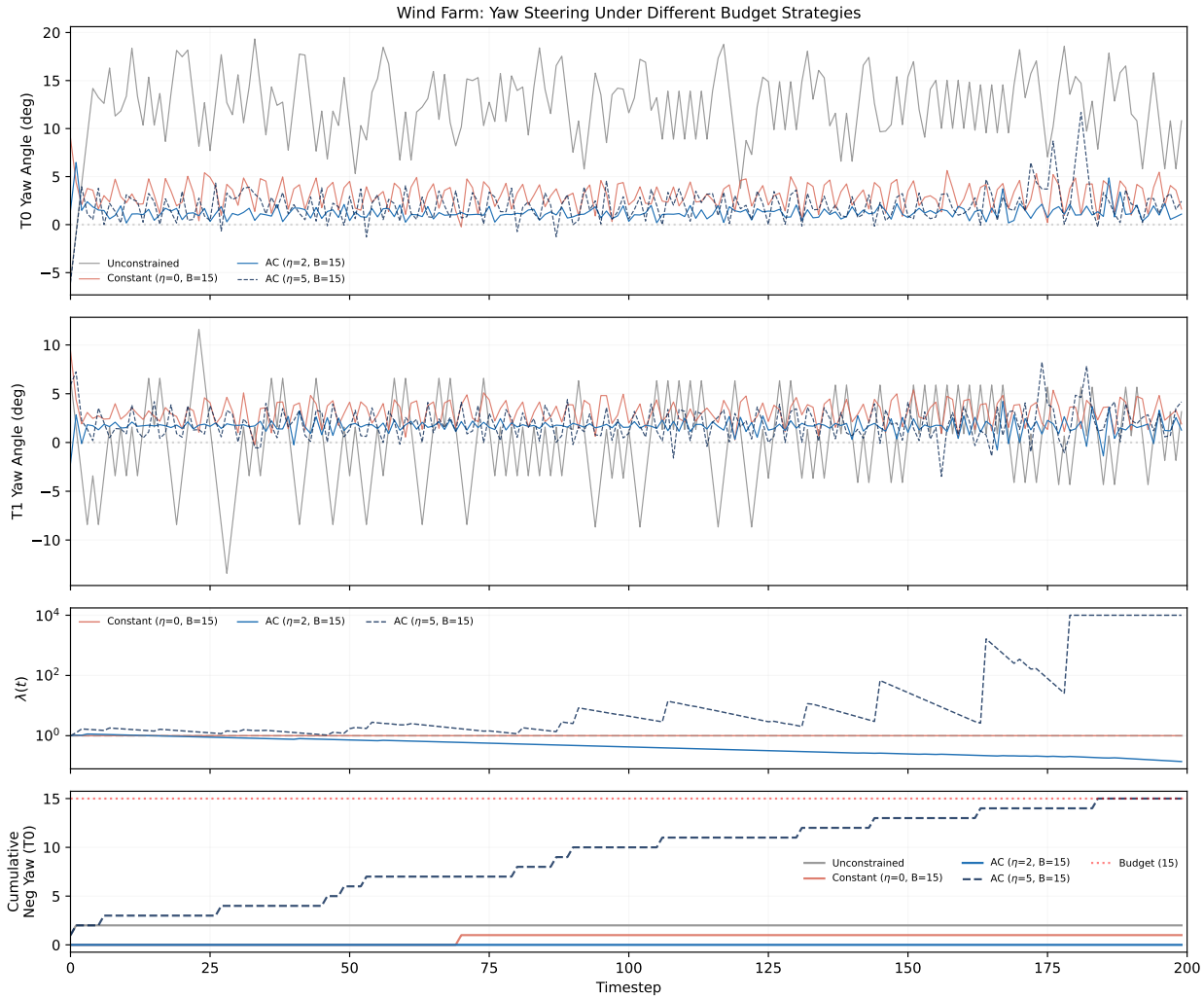

Figure 4: Wind farm yaw time series (one episode, budget= 15). $\lambda(t)$ adapts: $\eta = 5$ drops during surplus and spikes near depletion. Cumulative neg-yaw shows $\eta = 5$ uses the full budget.

A.7 Learned Cost Critic on Wind Farm

A.8 Wind Farm Spatial Allocation

A.9 Hard Guard Ablation

A.10 Schedule Comparison: $1/u^\eta$ vs. $\exp(\eta(1/u - 1))$

A.11 Budget Flexibility

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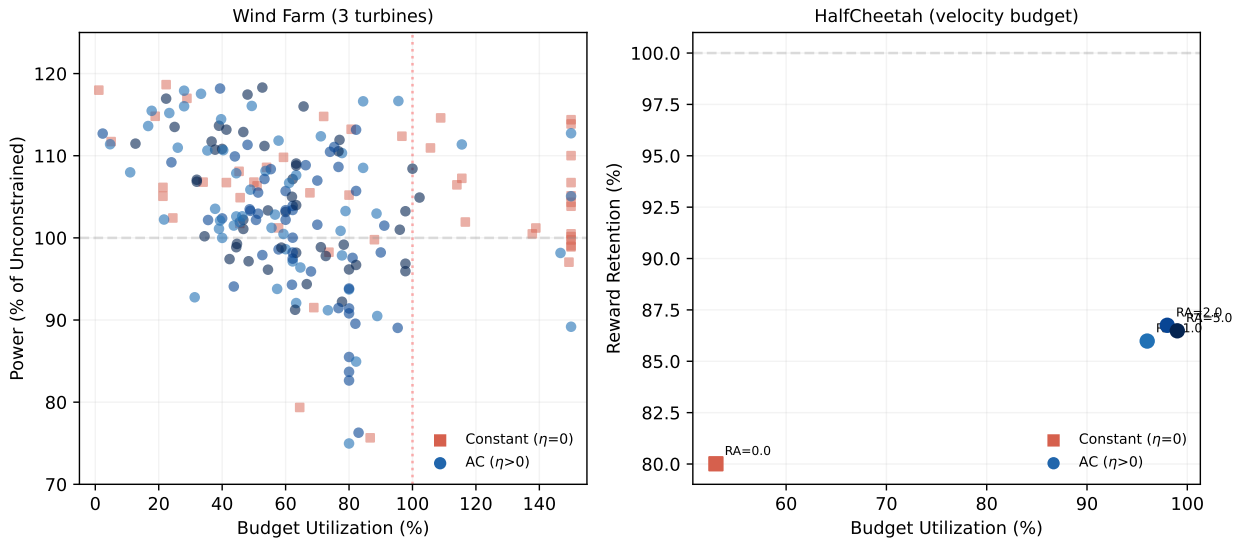

Figure 5: Budget utilization vs. reward retention. Constant penalty (red) cannot achieve both; AC schedule (blue) clusters in the efficient top-right region.

Table 4: Learned cost critic Q_c on 3-turbine wind farm (10 episodes). Q_c trained on 100 episodes with nonlinear fatigue cost. Power exceeding unconstrained reflects Q_c steering toward configurations that are both lower-fatigue and higher-power.

Method	Power (MW)	NegYaw	Fatigue	Fatigue ×
Unconstrained	24.6 ± 5.6	[84, 96, 94]	54.6	1.0×
Q_c, $\eta = 2$, gs=0.05	$\mathbf{28.1} \pm 2.7$	[43, 44, 43]	**9.5**	5.7×
Q_c, $\eta = 5$, gs=0.5	28.8 ± 2.0	[55, 54, 51]	10.4	5.3×
Analytical surr. $\eta = 5$	29.2	[80, 81, 83]	12.8	4.3×

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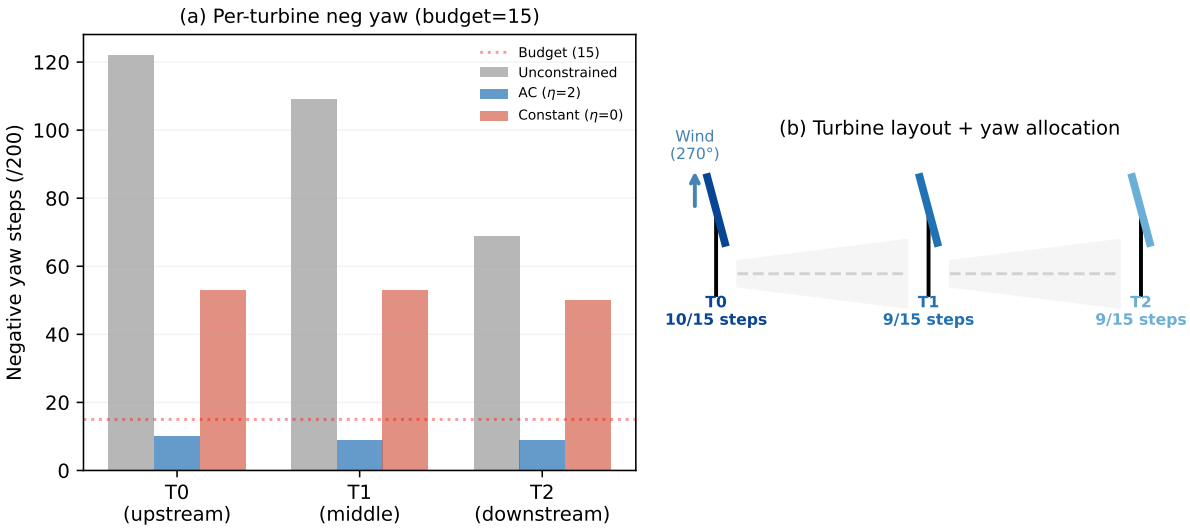

Figure 6: Wind farm spatial yaw allocation. (a) Unconstrained agent uses negative yaw heavily on all turbines; AC ($\eta = 2$) constrains to within the 15-step budget; constant penalty over-spends. (b) Turbine layout.

Table 5: Hard guard ablation (3-turbine wind farm, budget=15/200 steps, $\eta = 2$, $k = 2$, gs=0.1, 10 episodes).

Configuration	Power (MW)	NegYaw [T0, T1, T2]	Budget
Unconstrained	26.8 ± 3.9	[83, 87, 100]	—
Both (AC + guard)	24.7 ± 5.1	[13, 16, 19]	✓
AC only (no guard)	26.3 ± 3.0	[11, 11, 11]	✓
Guard only ($\eta = 0$)	24.8 ± 5.7	[29, 51, 58]	×

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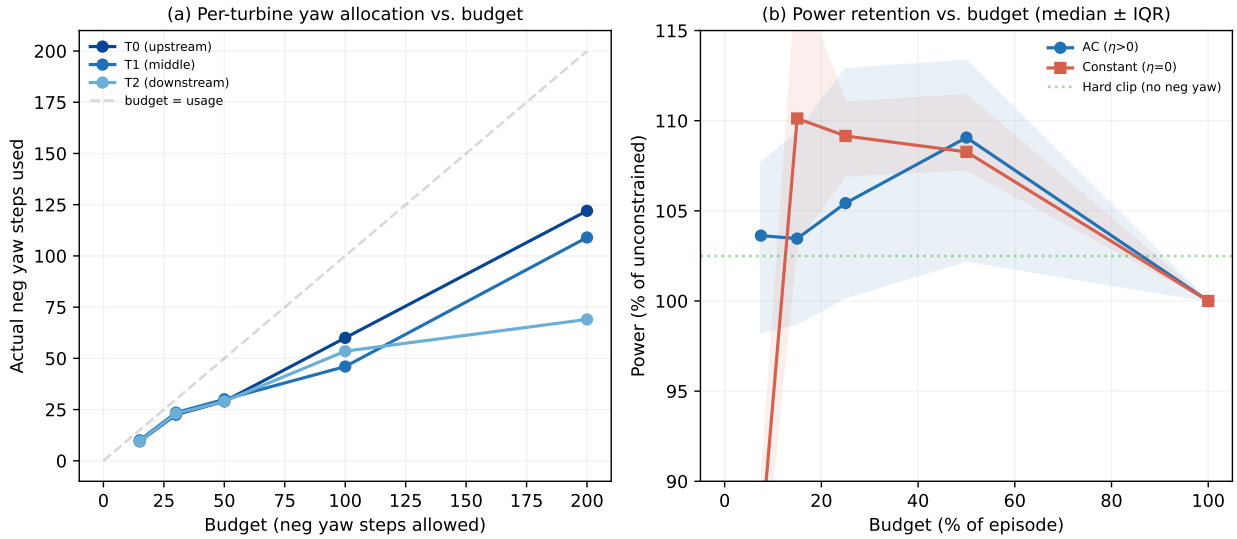

Figure 7: Wind farm budget effect. (a) Per-turbine allocation: as budget tightens, downstream T2 reduces negative yaw first. (b) Power retention remains within 5% of unconstrained.

Table 6: Schedule comparison on 3-turbine wind farm (budget=15/200 steps, $k = 2$, gs=0.1, 10 episodes).

Schedule	η	Power (MW)	NegYaw [T0, T1, T2]
$\exp(\eta(1/u - 1))$	0.5	24.3 ± 6.3	[32, 36, 37]
$\exp(\eta(1/u - 1))$	2.0	25.5 ± 5.3	[19, 24, 24]
$\exp(\eta(1/u - 1))$	5.0	27.8 ± 4.5	[14, 14, 14]
$u^{-\eta}$	0.5	24.9 ± 5.6	[56, 57, 58]
$u^{-\eta}$	2.0	24.8 ± 6.8	[29, 34, 35]
$u^{-\eta}$	5.0	26.6 ± 5.3	[22, 24, 24]

Table 7: Budget flexibility (3-turbine wind farm, $\eta = 2$, $k = 2$, gs=0.1, 10 episodes). A single policy serves arbitrary budget levels. Power retention is high from $d = 10$ onward.

Budget (steps/200)	Power (MW)	% Unconstrained	NegYaw [T0, T1, T2]
5 (2.5%)	26.4 ± 3.9	100.1%	[74, 76, 86]
10 (5%)	25.2 ± 5.3	95.7%	[25, 25, 26]
15 (7.5%)	29.2 ± 1.8	110.9%	[20, 20, 21]
25 (12.5%)	26.1 ± 4.1	99.1%	[17, 20, 20]
50 (25%)	26.2 ± 6.2	99.3%	[27, 39, 39]
75 (37.5%)	27.7 ± 3.6	105.2%	[54, 53, 62]
100 (50%)	22.3 ± 6.1	84.9%	[33, 50, 74]